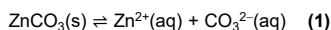


Effects on Solubility (Qualitative)

Consider a mixture containing a saturated solution of ZnCO_3 and undissolved ZnCO_3



1 Common ion effect

Explain how the addition of some $\text{Zn(NO}_3)_2(\text{s})$ affects the solubility of ZnCO_3

When some $\text{Zn(NO}_3)_2(\text{s})$ is added,

- $\text{Zn(NO}_3)_2$ dissolves and _____ increases.
- position of eqm (1) shifts _____
- hence, _____ $\text{ZnCO}_3(\text{s})$ is produced,
i.e. solubility of ZnCO_3 _____

2 Complex ion effect

Explain how the addition of excess aqueous NH_3 affects the solubility of ZnCO_3

When excess $\text{NH}_3(\text{aq})$ is added,

- $\text{Zn}^{2+}(\text{aq}) + 4\text{NH}_3(\text{aq}) \rightleftharpoons [\text{Zn(NH}_3)_4]^{2+}(\text{aq})$
- formation of the complex decreases _____
- position of eqm (1) shifts _____
- hence, _____ $\text{ZnCO}_3(\text{s})$ can be dissolved,
i.e. solubility of ZnCO_3 _____

3 Acid-base reaction

Explain how the addition of some $\text{HNO}_3(\text{aq})$ affects the solubility of ZnCO_3

When an acid is added,

- $\text{CO}_3^{2-}(\text{aq}) + \text{H}^+(\text{aq}) \rightleftharpoons \text{HCO}_3^-(\text{aq})$
- CO_3^{2-} reacts with H^+ and _____ decreases
- position of eqm (1) shifts _____
- hence, _____ $\text{ZnCO}_3(\text{s})$ can be dissolved,
i.e. solubility of ZnCO_3 _____

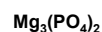
Writing K_{sp} expression

Write the K_{sp} expression for the following compounds and state the units.



$$K_{\text{sp}} \text{ of } \text{Ag}_3\text{PO}_4 =$$

units :



$$K_{\text{sp}} \text{ of } \text{Ca}_3(\text{PO}_4)_2 =$$

units :

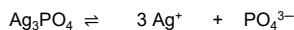
Calculating Solubility (Quantitative)

Pure water

The K_{sp} of Ag_3PO_4 is $8.9 \times 10^{-17} \text{ mol}^4 \text{ dm}^{-12}$

Calculate the solubility of Ag_3PO_4 in water

Let...



eqm conc/
 mol dm^{-3}

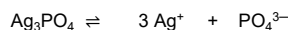
$$K_{\text{sp}} \text{ of } \text{Ag}_3\text{PO}_4 = \text{IP}$$

Common ion: PO_4^{3-}

The K_{sp} of Ag_3PO_4 is $8.9 \times 10^{-17} \text{ mol}^4 \text{ dm}^{-12}$

Calculate the solubility of Ag_3PO_4 in a solution which contains $1 \times 10^{-4} \text{ mol dm}^{-3}$ of PO_4^{3-}

Let...



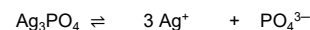
eqm conc/
 mol dm^{-3}

Common ion: Ag^+

The K_{sp} of Ag_3PO_4 is $8.9 \times 10^{-17} \text{ mol}^4 \text{ dm}^{-12}$

Calculate the solubility of Ag_3PO_4 in a solution which contains $1 \times 10^{-4} \text{ mol dm}^{-3}$ of Ag^+

Let...



eqm conc/
 mol dm^{-3}

Predicting Precipitation (Quantitative)

Key concepts

- Precipitation will occur when $IP > K_{sp}$
- After precipitation, the remaining solution is still a _____ solution, where $IP = K_{sp}$

Strategy: look out for key phrases to apply $IP = K_{sp}$

- "solubility"
- "saturated solution"
- "before it begins to precipitates"
- "mixture after precipitation"

solution X

500 cm³

Fig. 1

$$K_{sp} \text{ of } \text{Mg}_3(\text{PO}_4)_2 = 1.0 \times 10^{-24}$$

$$K_{sp} \text{ of } \text{Ag}_3\text{PO}_4 = 8.9 \times 10^{-17}$$

Solution X contains these ions:

$$[\text{Mg}^{2+}] = 2 \times 10^{-3} \text{ mol dm}^{-3}$$

$$[\text{Ag}^+] = 5 \times 10^{-4} \text{ mol dm}^{-3}$$

$$[\text{PO}_4^{3-}] = 1 \times 10^{-8} \text{ mol dm}^{-3}$$

Example 1

0.001 mol of solid $\text{Mg}(\text{NO}_3)_2$ was dissolved into 500 cm³ of solution X in Fig. 1.

Show that $\text{Mg}_3(\text{PO}_4)_2$ will precipitate out.

Example 3

Solid Na_3PO_4 was gradually dissolved into solution X in Fig. 1 until a precipitate is formed.

- Show that the precipitate is $\text{Mg}_3(\text{PO}_4)_2$, and not Ag_3PO_4

Example 2

Equal volumes of 0.002 mol dm⁻³ $\text{Mg}(\text{NO}_3)_2$ (aq) and solution X in Fig. 1 were mixed together.

Show that $\text{Mg}_3(\text{PO}_4)_2$ will not precipitate out.

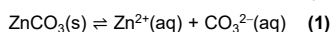
Solid Na_3PO_4 was continuously added into the solution until Ag_3PO_4 also precipitated out.

- Show that $[\text{Mg}^{2+}]$ in the solution is $1.25 \times 10^{-4} \text{ mol dm}^{-3}$ when Ag_3PO_4 precipitate began to form.

- Hence, calculate the amount of $\text{Mg}_3(\text{PO}_4)_2$ that had precipitated out from the solution before Ag_3PO_4 precipitate began to form.

Effects on Solubility (Qualitative)

Consider a mixture containing a saturated solution of ZnCO_3 and undissolved ZnCO_3



1 Common ion effect

Explain how the addition of some $\text{Zn}(\text{NO}_3)_2(\text{s})$ affects the solubility of ZnCO_3

When some $\text{Zn}(\text{NO}_3)_2(\text{s})$ is added,

- $\text{Zn}(\text{NO}_3)_2$ dissolves and $[\text{Zn}^{2+}]$ increases.
- position of eqm (1) shifts left
- hence, more $\text{ZnCO}_3(\text{s})$ is produced,
i.e. solubility of ZnCO_3 decreases

2 Complex ion effect

Explain how the addition of excess aqueous NH_3 affects the solubility of ZnCO_3

When excess $\text{NH}_3(\text{aq})$ is added,

- $\text{Zn}^{2+}(\text{aq}) + 4\text{NH}_3(\text{aq}) \rightleftharpoons [\text{Zn}(\text{NH}_3)_4]^{2+}(\text{aq})$
- formation of the complex decreases $[\text{Zn}^{2+}]$
- position of eqm (1) shifts right
- hence, more $\text{ZnCO}_3(\text{s})$ can be dissolved,
i.e. solubility of ZnCO_3 increases

3 Acid-base reaction

Explain how the addition of some $\text{HNO}_3(\text{aq})$ affects the solubility of ZnCO_3

When an acid is added,

- $\text{CO}_3^{2-}(\text{aq}) + \text{H}^+(\text{aq}) \rightleftharpoons \text{HCO}_3^-(\text{aq})$
- CO_3^{2-} reacts with H^+ and $[\text{CO}_3^{2-}]$ decreases
- position of eqm (1) shifts right
- hence, more $\text{ZnCO}_3(\text{s})$ can be dissolved,
i.e. solubility of ZnCO_3 increases

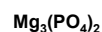
Writing K_{sp} expression

Write the K_{sp} expression for the following compounds and state the units.



$$K_{\text{sp}} \text{ of } \text{Ag}_3\text{PO}_4 = [\text{Ag}^+]^3 [\text{PO}_4^{3-}]$$

units : $\text{mol}^4 \text{dm}^{-12}$



$$K_{\text{sp}} \text{ of } \text{Ca}_3(\text{PO}_4)_2 = [\text{Mg}^{2+}]^3 [\text{PO}_4^{3-}]^2$$

units : $\text{mol}^5 \text{dm}^{-15}$

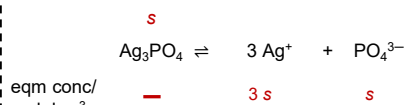
Calculating Solubility (Quantitative)

Pure water

The K_{sp} of Ag_3PO_4 is $8.9 \times 10^{-17} \text{ mol}^4 \text{dm}^{-12}$

Calculate the solubility of Ag_3PO_4 in water

Let... solubility of Ag_3PO_4 be s



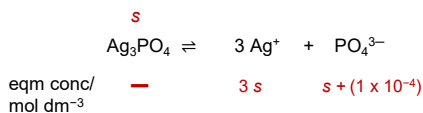
$$\begin{aligned} K_{\text{sp}} \text{ of } \text{Ag}_3\text{PO}_4 &= \text{IP (saturated solution)} \\ 8.9 \times 10^{-17} &= [\text{Ag}^+]^3 [\text{PO}_4^{3-}] \\ &= [3s]^3 [s] \\ 8.9 \times 10^{-17} &= 27s^4 \\ s &= 4.26 \times 10^{-5} \text{ mol dm}^{-3} \end{aligned}$$

Common ion: PO_4^{3-}

The K_{sp} of Ag_3PO_4 is $8.9 \times 10^{-17} \text{ mol}^4 \text{dm}^{-12}$

Calculate the solubility of Ag_3PO_4 in a solution which contains $1 \times 10^{-4} \text{ mol dm}^{-3}$ of PO_4^{3-}

Let... solubility of Ag_3PO_4 be s



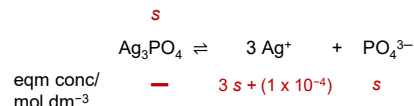
$$\begin{aligned} K_{\text{sp}} \text{ of } \text{Ag}_3\text{PO}_4 &= \text{IP (saturated solution)} \\ 8.9 \times 10^{-17} &= [\text{Ag}^+]^3 [\text{PO}_4^{3-}] \\ &= [3s]^3 [s + (1 \times 10^{-4})] \\ 8.9 \times 10^{-17} &\approx [3s]^3 [(1 \times 10^{-4})] \quad (\text{since } s \text{ small}) \\ s &= 3.21 \times 10^{-5} \text{ mol dm}^{-3} \end{aligned}$$

Common ion: Ag^+

The K_{sp} of Ag_3PO_4 is $8.9 \times 10^{-17} \text{ mol}^4 \text{dm}^{-12}$

Calculate the solubility of Ag_3PO_4 in a solution which contains $1 \times 10^{-4} \text{ mol dm}^{-3}$ of Ag^+

Let... solubility of Ag_3PO_4 be s



$$\begin{aligned} K_{\text{sp}} \text{ of } \text{Ag}_3\text{PO}_4 &= \text{IP (saturated solution)} \\ 8.9 \times 10^{-17} &= [\text{Ag}^+]^3 [\text{PO}_4^{3-}] \\ &= [3s + (1 \times 10^{-4})]^3 [s] \\ 8.9 \times 10^{-17} &\approx [1 \times 10^{-4}]^3 [s] \quad (\text{since } s \text{ small}) \\ s &= 8.90 \times 10^{-5} \text{ mol dm}^{-3} \end{aligned}$$

Predicting Precipitation (Quantitative)

Key concepts

- Precipitation will occur when $IP > K_{sp}$
- After precipitation, the remaining solution is still a saturated solution, where $IP = K_{sp}$

Strategy: look out for key phrases to apply $IP = K_{sp}$

- "solubility"
- "saturated solution"
- "before it begins to precipitates"
- "mixture after precipitation"

solution X
500 cm³

Fig. 1

K_{sp} of $Mg_3(PO_4)_2 = 1.0 \times 10^{-24}$
 K_{sp} of $Ag_3PO_4 = 8.9 \times 10^{-17}$

Solution X contains these ions:
 $[Mg^{2+}] = 2 \times 10^{-3} \text{ mol dm}^{-3}$
 $[Ag^+] = 5 \times 10^{-4} \text{ mol dm}^{-3}$
 $[PO_4^{3-}] = 1 \times 10^{-8} \text{ mol dm}^{-3}$

Example 1

0.001 mol of solid $Mg(NO_3)_2$ was dissolved into 500 cm³ of solution X in Fig. 1. ~ common ion effect

Show that $Mg_3(PO_4)_2$ will precipitate out.

Show that $IP > K_{sp}$ of $Mg_3(PO_4)_2$

$$[Mg^{2+}]_{\text{upon mixing}} = \frac{0.001}{\left(\frac{500}{1000}\right)} + (2 \times 10^{-3}) = 4 \times 10^{-3} \text{ mol dm}^{-3}$$

$$[PO_4^{3-}]_{\text{upon mixing}} = 1 \times 10^{-8} \text{ mol dm}^{-3}$$

$$\begin{aligned} IP &= [Mg^{2+}]^3 [PO_4^{3-}]^2 \\ &= (4 \times 10^{-3})^3 (1 \times 10^{-8})^2 \\ &= (4 \times 10^{-3})^3 (1 \times 10^{-8})^2 \\ &= 6.4 \times 10^{-24} > K_{sp} \text{ of } Mg_3(PO_4)_2 \end{aligned}$$

Since ionic product $> K_{sp}$, $Mg_3(PO_4)_2$ will ppt out

Example 2

Equal volumes of 0.002 mol dm⁻³ $Mg(NO_3)_2$ (aq) and solution X in Fig. 1 were mixed together. ~ common ion effect
~ dilution

Show that $Mg_3(PO_4)_2$ will not precipitate out.

Show that $IP < K_{sp}$ of $Mg_3(PO_4)_2$

$$\begin{aligned} [Mg^{2+}]_{\text{upon mixing}} &= \left(\frac{0.002}{2}\right) + \left(\frac{2 \times 10^{-3}}{2}\right) \\ &= 2 \times 10^{-3} \text{ mol dm}^{-3} \end{aligned}$$

$$\begin{aligned} [PO_4^{3-}]_{\text{upon mixing}} &= \left(\frac{1 \times 10^{-8}}{2}\right) \\ &= 5 \times 10^{-9} \text{ mol dm}^{-3} \end{aligned}$$

$$\begin{aligned} IP &= [Mg^{2+}]^3 [PO_4^{3-}]^2 \\ &= (2 \times 10^{-3})^3 (5 \times 10^{-9})^2 \\ &= 2 \times 10^{-25} < K_{sp} \text{ of } Mg_3(PO_4)_2 \end{aligned}$$

Since ionic product $< K_{sp}$, $Mg_3(PO_4)_2$ will not ppt out

Example 3

Solid Na_3PO_4 was gradually dissolved into solution X in Fig. 1 until a precipitate is formed.

- Show that the precipitate is $Mg_3(PO_4)_2$, and not Ag_3PO_4

Show that $IP = K_{sp}$ of $Mg_3(PO_4)_2$ occurs before $IP = K_{sp}$ of Ag_3PO_4

$$\begin{aligned} K_{sp} \text{ of } Mg_3(PO_4)_2 &= IP \\ 1.0 \times 10^{-24} &= [Mg^{2+}]^3 [PO_4^{3-}]^2 \\ &= [2 \times 10^{-3}]^3 [PO_4^{3-}]^2 \\ \text{min. } [PO_4^{3-}] &= \sqrt{\frac{(1.0 \times 10^{-24})}{(2 \times 10^{-3})^3}} \\ &= 1.12 \times 10^{-8} \text{ mol dm}^{-3} \end{aligned}$$

$$\begin{aligned} K_{sp} \text{ of } Ag_3PO_4 &= IP \\ 8.9 \times 10^{-17} &= [Ag^+]^3 [PO_4^{3-}] \\ &= [5 \times 10^{-4}]^3 [PO_4^{3-}] \\ \text{min } [PO_4^{3-}] &= \frac{(8.9 \times 10^{-17})}{(5 \times 10^{-4})^3} \\ &= 7.12 \times 10^{-7} \text{ mol dm}^{-3} \end{aligned}$$

Since lower $[PO_4^{3-}]$ is needed to ppt $Mg_3(PO_4)_2$; hence it ppt out first

Solid Na_3PO_4 was continuously added into the solution until Ag_3PO_4 also precipitated out.

- Show that $[Mg^{2+}]$ in the solution is $1.25 \times 10^{-4} \text{ mol dm}^{-3}$ when Ag_3PO_4 precipitate began to form. ~ solution remains saturated after precipitation

When Ag_3PO_4 ppt was first formed,

- $IP = K_{sp}$ of Ag_3PO_4
- $IP = K_{sp}$ of $Mg_3(PO_4)_2$ (still a saturated solution)

$$\begin{aligned} K_{sp} \text{ of } Ag_3PO_4 &= IP \\ 8.9 \times 10^{-17} &= [Ag^+]^3 [PO_4^{3-}] \\ &= [5 \times 10^{-4}]^3 [PO_4^{3-}] \\ \text{min } [PO_4^{3-}] &= 7.12 \times 10^{-7} \text{ mol dm}^{-3} \end{aligned}$$

$$\begin{aligned} K_{sp} \text{ of } Mg_3(PO_4)_2 &= IP \\ 1.0 \times 10^{-24} &= [Mg^{2+}]^3 [PO_4^{3-}]^2 \\ &= [Mg^{2+}]^3 [7.12 \times 10^{-7}]^2 \\ [Mg^{2+}] &= 1.25 \times 10^{-4} \text{ mol dm}^{-3} \end{aligned}$$

- Hence, calculate the amount of $Mg_3(PO_4)_2$ that had precipitated out from the solution before Ag_3PO_4 precipitate began to form. ~ conservation of mass

$$\begin{aligned} \text{total } n_{Mg^{2+}} &= n_{Mg^{2+}} \text{ in solution} + n_{Mg^{2+}} \text{ in precipitate} \\ (2 \times 10^{-3})\left(\frac{500}{1000}\right) &= (1.25 \times 10^{-4})\left(\frac{500}{1000}\right) + n_{Mg^{2+}} \text{ in precipitate} \\ n_{Mg^{2+}} \text{ in precipitate} &= 9.38 \times 10^{-4} \text{ mol} \\ n_{Mg_3(PO_4)_2} \text{ precipitate} &= \left(\frac{1}{3}\right)(9.38 \times 10^{-4}) = 3.13 \times 10^{-4} \text{ mol} \end{aligned}$$